In [6]:
```
import pandas as pd

df = pd.read_csv('customer_shopping_behavior.csv')
```

In [7]:
```
df.head()
```

Out[7]:

Columns 1–8 of 19

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location
0	1	55	Male	Blouse	Clothing	53	Kentucky
1	2	19	Male	Sweater	Clothing	64	Maine
2	3	50	Male	Jeans	Clothing	73	Massachusetts
3	4	21	Male	Sandals	Footwear	90	Rhode Island
4	5	45	Male	Blouse	Clothing	49	Oregon

Columns 9–15 of 19 · first column repeated

	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
0	L	Gray	Winter	3.1	Yes	Express	Yes
1	L	Maroon	Winter	3.1	Yes	Express	Yes
2	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes
3	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes
4	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes

Columns 16–19 of 19 · first column repeated

	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	Yes	14	Venmo	Fortnightly
1	Yes	2	Cash	Fortnightly
2	Yes	23	Credit Card	Weekly
3	Yes	49	PayPal	Weekly
4	Yes	31	PayPal	Annually

In [8]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   Customer ID             3900 non-null   int64  
 1   Age                     3900 non-null   int64  
 2   Gender                  3900 non-null   str    
 3   Item Purchased          3900 non-null   str    
 4   Category                3900 non-null   str    
 5   Purchase Amount (USD)   3900 non-null   int64  
 6   Location                3900 non-null   str    
 7   Size                    3900 non-null   str    
 8   Color                   3900 non-null   str    
 9   Season                  3900 non-null   str    
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   str    
 12  Shipping Type           3900 non-null   str    
 13  Discount Applied        3900 non-null   str    
 14  Promo Code Used         3900 non-null   str    
 15  Previous Purchases      3900 non-null   int64  
 16  Payment Method          3900 non-null   str    
 17  Frequency of Purchases  3900 non-null   str    
dtypes: float64(1), int64(4), str(13)
memory usage: 835.9 KB
```

In [9]:
```
df.describe(include='all')
```

Out[9]: Columns 1–8 of 19

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900
unique	NaN	NaN	2	25	4	NaN	50
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana
freq	NaN	NaN	2652	171	1737	NaN	96
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN

Columns 9–15 of 19 · first column repeated

	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied
count	3900	3900	3900	3863.000000	3900	3900	3900
unique	4	25	4	NaN	2	6	2
top	M	Olive	Spring	NaN	No	Free Shipping	No
freq	1755	177	999	NaN	2847	675	2223
mean	NaN	NaN	NaN	3.750065	NaN	NaN	NaN
std	NaN	NaN	NaN	0.716983	NaN	NaN	NaN
min	NaN	NaN	NaN	2.500000	NaN	NaN	NaN
25%	NaN	NaN	NaN	3.100000	NaN	NaN	NaN
50%	NaN	NaN	NaN	3.800000	NaN	NaN	NaN
75%	NaN	NaN	NaN	4.400000	NaN	NaN	NaN
max	NaN	NaN	NaN	5.000000	NaN	NaN	NaN

Columns 16–19 of 19 · first column repeated

	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900	3900.000000	3900	3900
unique	2	NaN	6	7

	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
top	No	NaN	PayPal	Every 3 Months
freq	2223	NaN	677	584
mean	NaN	25.351538	NaN	NaN
std	NaN	14.447125	NaN	NaN
min	NaN	1.000000	NaN	NaN
25%	NaN	13.000000	NaN	NaN
50%	NaN	25.000000	NaN	NaN
75%	NaN	38.000000	NaN	NaN
max	NaN	50.000000	NaN	NaN

In [10]:
```
df.isnull().sum()
```

Out[10]:
```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

In [11]:
```
df['Review Rating'] = df.groupby('Category')['Review
Rating'].transform(lambda x: x.fillna(x.median()))
```

In [12]:
```
df.isnull().sum()
```

Out[12]:
```
Customer ID               0
Age                       0
Gender                    0
Item Purchased            0
Category                  0
Purchase Amount (USD)     0
Location                  0
Size                      0
Color                     0
Season                    0
Review Rating             0
Subscription Status       0
Shipping Type             0
Discount Applied          0
Promo Code Used           0
Previous Purchases        0
Payment Method            0
Frequency of Purchases    0
dtype: int64
```

In [13]:
```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df = df.rename(columns={'purchase_amount_(usd)':'purchase_amount'})
```

In [14]:
```
df.columns
```

Out[14]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

In [15]:
```
# create a column age_group
labels = ['Young Adult','Adult','Middle-age','Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)
```

In [16]:
```
df[['age','age_group']].head(10)
```

Out[16]:

	age	age_group
0	55	Middle-age
1	19	Young Adult
2	50	Middle-age
3	21	Young Adult
4	45	Middle-age
5	46	Middle-age
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-age

In [17]:
```
#creat column purchase_frequency_days

frequency_mapping = {
    'Fortnightly' : 14,
    'Weekly' : 7,
    'Monthly' : 30,
    'Quarterly' : 90,
    'Bi-Weekly' : 14,
    'Annually' : 365,
    'Every 3 months' : 90
}

df['purchase_frequency_days'] =
df['frequency_of_purchases'].map(frequency_mapping)
```

In [18]:
```
df[['purchase_frequency_days','frequency_of_purchases']].head(10)
```

Out[18]:

	purchase_frequency_days	frequency_of_purchases
0	14.0	Fortnightly
1	14.0	Fortnightly
2	7.0	Weekly
3	7.0	Weekly
4	365.0	Annually
5	7.0	Weekly
6	90.0	Quarterly
7	7.0	Weekly
8	365.0	Annually
9	90.0	Quarterly

In [19]:
```
df[['discount_applied','promo_code_used']].head(10)
```

Out[19]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

In [20]:
```
(df['discount_applied'] == df['promo_code_used']).all()
```

Out[20]:
```
np.True_
```

In [21]:
```
df = df.drop('promo_code_used', axis=1)
```

In [22]:
```
df.columns
```

Out[22]:
```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'previous_purchases', 'payment_method',
       'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='str')
```

In [23]:
```
pip install psycopg2-binary sqlalchemy
```

```
Requirement already satisfied: psycopg2-binary in
C:\Users\lenovo\anaconda3\Lib\site-packages (2.9.13)
Requirement already satisfied: sqlalchemy in
C:\Users\lenovo\anaconda3\Lib\site-packages (2.0.51)
Requirement already satisfied: greenlet ⩾ 1 in
C:\Users\lenovo\anaconda3\Lib\site-packages (from sqlalchemy) (3.3.2)
Requirement already satisfied: typing-extensions ⩾ 4.6.0 in
C:\Users\lenovo\AppData\Roaming\Python\Python314\site-packages (from
sqlalchemy) (4.16.0)
Note: you may need to restart the kernel to use updated packages.
```

In [24]:
```
from sqlalchemy import create_engine

username = 'postgres'
password = 'lordjesus7$'
host = 'localhost'
port = '5432'
database = "customer_behavior"

engine = create_engine(
    f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}"
)

table_name = "customer"

df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"Data successfully loaded into table '{table_name}' in database
'{database}'.")
```

```
Data successfully loaded into table 'customer' in database
'customer_behavior'.
```